



Genetic characteristics and diversity of PDC variants of *Pseudomonas aeruginosa* and its clinical relevance

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ABSTRACT

Pseudomonas aeruginosa exhibits significant antibiotic resistance facilitated by both intrinsic and acquired mechanisms, prominently through *Pseudomonas*-derived cephalosporinase (PDC), serine Ambler class C β-lactamases encoded by the AmpC. AmpC, involved in the peptidoglycan recycling pathway, is regulated by genes such as ampD, ampR, and ampG, leading to increased expression and resistance to various beta-lactams. PDCs are classified into three main types: classical class C β-lactamases, extended-spectrum class C β-lactamases (ESAC β-lactamases), and inhibitor-resistant class C β-lactamases. This study aimed to identify prevalent PDC variants and its genetic characteristics in Indian and global *P. aeruginosa* isolates, focusing on their role in β-lactam resistance. Analyzing PDC sequences from 111 *P. aeruginosa* isolates collected at Christian Medical College (CMC), Vellore, we found the ESAC allele PDC-447 to be the most widespread among Indian isolates, present in 18 % of carbapenem-resistant and 11 % of carbapenem-susceptible strains. Global and Indian isolates PDC variants were validated using the NCBI PathogenWatch database, and the sequenced PDC region compared to PDC-1. PDC-398 and PDC-397 followed in prevalence among carbapenem-resistant isolates, while PDC-5 (ESAC) and PDC-1 (classical class C) were common in carbapenem-susceptible strains. A global analysis of 19,478 genomes revealed significant prevalence of ESAC variants such as PDC-3 (17.28 %) and PDC-5 (12.91 %), alongside classical class C beta-lactamases like PDC-8 (10.65 %). Indian isolates exhibited distinct patterns with PDC-3 and PDC-5 prevailing at 19.84 % and 10 %, respectively. Mutations in the omega loop, H-helix, and R2 region of PDCs were linked to enhanced antibiotic resistance, particularly the T105A mutation in the H-helix region. These findings underscore the complexity of antimicrobial resistance mechanisms in *P. aeruginosa* and highlight the need for novel therapeutic strategies and continuous surveillance to manage infections by this versatile pathogen. Understanding the prevalence and genetic characteristics of PDC variants is crucial for effective treatment strategies against *P. aeruginosa* and combating antibiotic resistance.

1. Introduction

Pseudomonas aeruginosa exhibits significant resistance to antibiotics through intrinsic and acquired mechanisms. One key intrinsic resistance mechanism involves the *Pseudomonas*-derived cephalosporinases (PDCs), which are serine β-lactamases encoded by the chromosomal gene AmpC (Langendonk et al., 2021; Saleh et al., 2024). These

enzymes, belonging to Ambler class C β-lactamase, confer inherent resistance to 4th generation cephalosporin antibiotics (Cabot et al., 2016; Colque et al., 2022; Sadeghi-Deylamdeh and Jafari-Sales, 2021). There are three main classes of PDCs: classical class C β-lactamase, extended-spectrum class C β-lactamase (ESAC β-lactamase), and inhibitor-resistant class C β-lactamase. Classical PDCs provide resistance up to 2nd generation cephalosporins, while ESAC β-lactamases and

Abbreviations: AST, Antimicrobial Susceptibility Testing; PDC, *Pseudomonas*-Derived Cephalosporinase; ESBL, Extended Spectrum Beta-lactamase; ESAC, Extended Spectrum AmpC; MBL, Metallo-beta-lactamase; CRPA/CSPA, Carbapenem Resistant *P. aeruginosa*, Carbapenem Susceptible *P. aeruginosa*; NCBI, National Center for Biotechnology Information; CMC, Christian Medical College.

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inhibitor-resistant PDCs broaden the activity spectrum, including resistance to 3rd and 4th generation cephalosporins and beta-lactamase inhibitors. This diversity in PDCs highlights the severity of antimicrobial resistance with *Pseudomonas* biofilm, PDC variants can confer resistance to anti-pseudomonal cephalosporins which is deployed as a first-line therapy and inhibitor-resistant PDC can confer resistance to novel BL/BLI combinations in *P. aeruginosa*, making it challenging to combat infections caused by this bacterium (Berrazeg et al., 2015; Nordmann and Mammeri, 2007; Al-Qaysi et al., 2021).

Structurally, the enzyme consists of two domains: an α domain and an α/β domain situated on opposite sides of the molecule. The active site, which contains essential R1 and R2 side chains, along with the Ω -loop and H-10, H-11 helices, is positioned at the center between both domains (Jeon et al., 2015). The Ω -loop, crucial for enzyme function, form hydrogen bond interactions with the H-2 helix near Ser64, a critical catalytic site enzyme (Jacoby, 2009; Nordmann and Mammeri, 2007) (Fig. 2 (a, b)). The amino acid region spanning from 178 to 226, encompassing the Ω -loop and part of the substrate-binding pocket, is highly conserved to mutations. Mutations in this region have a significant impact on the enzyme's activity spectrum. For instance, mutations may extend the Ω -loop or induce conformational changes, widening the active site. This alteration allows the enzyme to accommodate larger side chains of β -lactam antibiotics and facilitates their hydrolysis. This structural insight helps explain how mutations in specific regions can influence the enzyme's ability to interact with and degrade antibiotics, contributing to antibiotic resistance (Berrazeg et al., 2015; Nordmann and Mammeri, 2007).

Anti-pseudomonal agents are effective against *P. aeruginosa* due to their low susceptibility to chromosomal beta-lactamase produced by the bacterium. However, many beta-lactamase inhibitors (BLIs) lack coverage against PDC inhibition. Novel combinations such as ceftazidime/avibactam, ceftolozane/tazobactam, meropenem/vaborbactam, and cefepime/zidebactam employ diverse mechanisms to target *P. aeruginosa*, offering promising alternatives (Bhowmick and Weinstein, 2020; Moya et al., 2020). Avibactam, a beta-lactamase inhibitor, exhibits a spectrum that includes class C enzymes, protection its partner beta-lactam against PDC hydrolysis (Li et al., 2015). Ceftolozane's activity against *P. aeruginosa* remains largely modest by tazobactam (Moya et al., 2010). Zidebactam moderately inhibits class C beta-lactamase but primarily enhances beta-lactam (Moya et al., 2017; Papp-Wallace et al., 2018). Vaborbactam, a cyclic boronic-based BLI, does not offer significant protection against PDC hyperexpression but enhances meropenem's efficacy against *P. aeruginosa* (Bhowmick and Weinstein, 2020). Notably, clinically significant BLIs like avibactam, clavulanic acid, tazobactam, and sulbactam have been associated with the induction of PDC expression in various studies, indicating the complexity of antimicrobial resistance mechanisms (Weber and Sanders, 1990; Lister et al., 1999; Livermore et al., 2017). Glucose, galactose and the combination of ciprofloxacin is the enhancing of *P. aeruginosa* biofilm and increase survival strategy against ciprofloxacin.

In this study, we aimed to decipher the prevalent variants of PDC and determine their occurrence in both Indian isolates and globally. To achieve this, we analyzed PDC sequences amplified from carbapenem-resistant and susceptible *P. aeruginosa* isolates collected from Christian Medical College (CMC), Vellore and, referenced PDC variants from global *P. aeruginosa* genomes in the NCBI database. This collated investigation sheds light on the importance of PDC variants and its clinical relevance.

2. Materials and methods

2.1. Isolates collection

A total of 111 *P. aeruginosa* isolates were collected from blood and endotracheal aspirate samples of patients at Christian Medical College, Vellore, in 2023. Antimicrobial susceptibility testing was conducted

according to the CLSI guidelines. Standardized sterile media, including Mueller-Hinton agar (MHA), was used for *P. aeruginosa* organisms. Bacterial suspensions were prepared and adjusted to match a McFarland 0.5 turbidity standard. Disks containing commercially prepared antimicrobial agents were placed on inoculated plates. The panel of antibiotics included ceftazidime, cefepime, aztreonam, piperacillin/tazobactam, cefetizole/sulbactam, imipenem, meropenem, tobramycin, amikacin, gentamicin, and ciprofloxacin (Jekarl et al., 2010). The plates were then incubated at $35 \pm 2^\circ\text{C}$. Zone diameters of inhibition were measured after 16–18 h and interpreted using CLSI M100-S24 standards to classify organisms as susceptible, intermediate, or resistant (Murshid et al., 2024). Based on the results, the isolates were categorized as carbapenem-resistant ($n = 50$) and carbapenem-susceptible ($n = 61$). Quality control was maintained using reference strains to ensure test accuracy and reproducibility.

2.2. PDC amplification and sequencing

Bacterial isolates of *P. aeruginosa* were cultured as pure cultures on MacConkey agar plates and allowed to incubate for 24 h under conditions to *P. aeruginosa* growth. Genomic DNA extraction from the chosen bacterial isolates was conducted using the QIAamp DNA Mini Kit from QIAGEN India Pvt. Ltd., the extraction process was performed to yield high-quality DNA samples. Subsequently, the concentration and purity of the extracted DNA were checked using Nano-Drop One spectrophotometry. The extracted DNA was then stored at -20°C to maintain its integrity until further analysis. PCR amplification targeting the PDC gene, 1295 bp in length, was carried out under optimized conditions, including an annealing temperature of 58°C using specific primers PDC-F (5'-CGGTTTCTCATGCAGCCAAC-3') and PDC-R (5'-ACCATCATAGCCAGGACCG-3'). (Satlin et al., 2023). Gel electrophoresis was employed to visualize the amplified PCR products, confirming the presence of the expected DNA band size indicative of the PDC presence.

2.3. Sanger sequencing and identification of PDC variants

Samples yielding positive PCR results were selected for Sanger sequencing to determine the nucleotide sequence of the amplified PDC gene fragment accurately. The PDC gene of *P. aeruginosa* was amplified using above mentioned primers. The complete PDC gene was sequenced, with both forward and reverse DNA strands were sequenced. Nucleotide sequences using both forward and reverse primers were acquired through Sanger sequencing using 3500 series genetic analyzer. The sequences were merged using the Emboss merger tool available online. The resulting sequences in FASTA format were converted to protein sequences and analyzed through BLAST against the reference PDC-1 sequence in NCBI (<https://www.ncbi.nlm.nih.gov/>). This analysis aimed to identify PDC variants curated within the NCBI databases for the assessment of genetic diversity, identification of mutations, and comparison with PDC-1 reference database (NCBI reference sequence: NG_049865.1).

2.4. Mutation identification

Predominant PDC variants, displaying occurrences exceeding 10 % or approximately 10 % among the total 111 CRPA and CSPA isolates, underwent further analysis for amino acid substitution, frameshift mutations in their sequences. These analyses were conducted with reference to PDC-1, identified as the wild-type PDC. The online PDC database maintained by Oliver A (<https://arpbigdisba.com/pseudomonas-aeruginosa-derived-cephalosporinase-pdc-database/>) was used for the identification of mutations.

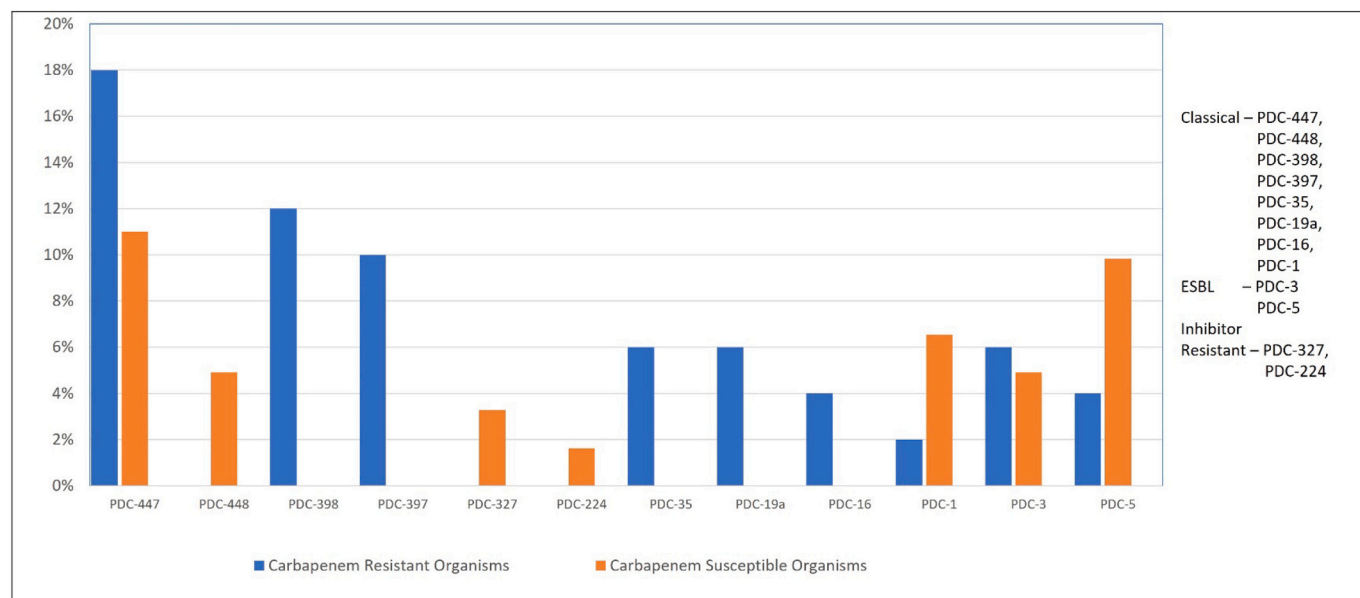
2.5. Global and Indian data analysis

22,133 *P. aeruginosa* genomes were curated from NCBI pathogen

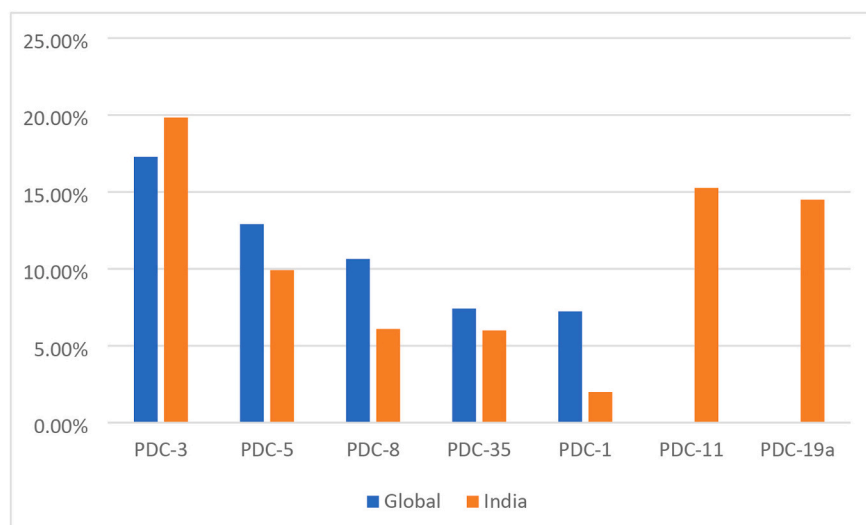
database (<https://www.ncbi.nlm.nih.gov/pathogens/isolates>). The existence of various PDC alleles was examined using the antimicrobial resistance genotype data of these genomes. After eliminating genomes containing partial and mis-translated genes, only the full-length PDC gene-carrying genomes (19,478 genomes) remained for analysis. Among these examined genomes, 235 distinct PDC alleles were identified. Similarly, 131 Indian *P. aeruginosa* genomes were retrieved from NCBI and examined for variants of the PDC. From this data, inferences have been made regarding the frequency, abundance, and spatial distribution of alleles.

3. Results

Out of the 111 *P. aeruginosa* isolates examined, the PDC-447 allele, identified as an ESAC (Extended-Spectrum AmpC Cephalosporinase) class C ampC beta-lactamase, is notably widespread. Specifically, it is present in 18 % of the carbapenem-resistant isolates and 11 % of the carbapenem-susceptible isolates. Among the carbapenem-resistant isolates ($n = 50$), PDC-398 and PDC-397 were the second and third most prevalent alleles attributing to 12 % and 10 %, respectively. Subsequently, PDC-3 and PDC-5 account for 6 % and 4 %, respectively. Fig. 1 (a). While among the carbapenem-susceptible isolates ($n = 61$), PDC-5 (ESAC) and PDC-1 (classical class C) have prevalence of 10 % and 7 %, respectively. Inhibitor resistance variants like PDC-327 and PDC-224



(a)



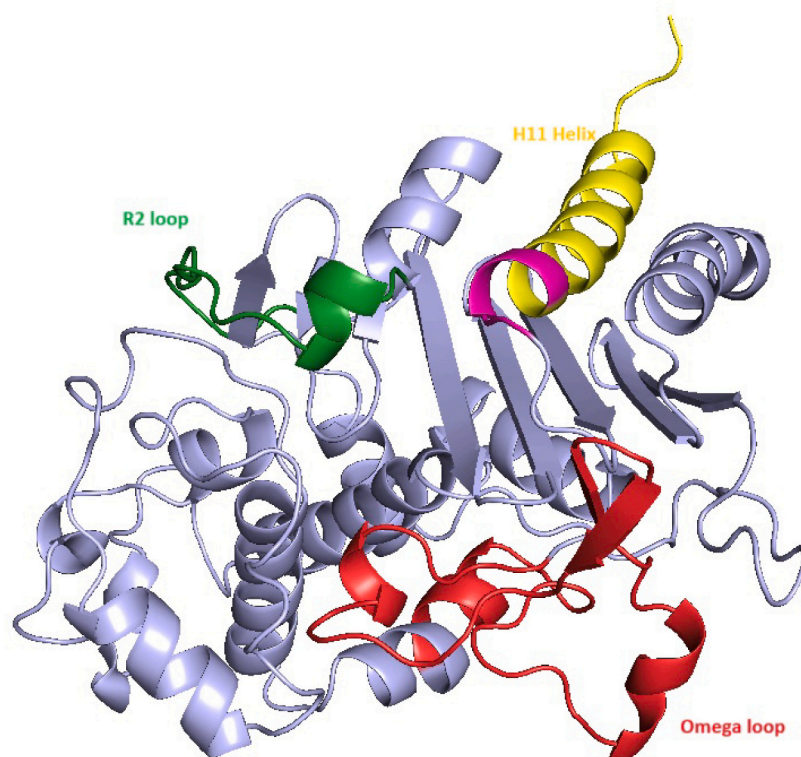
(b)

Fig. 1. (a) Distribution of PDC Variants in Carbapenem-Resistant and Susceptible Isolates of *P. aeruginosa*. (b) PDC variants distribution in India and globally.

were present at 3 % and 2 %, respectively. Overall, classical C class beta-lactamases are 90 % and 80 % in carbapenem-resistant and susceptible isolates, respectively. ESAC C class beta-lactamases are 10 % and 14.75 % in carbapenem-resistant and susceptible isolates. Inhibitor-resistant variants do not occur in carbapenem-resistant isolates but are present in 4.9 % of carbapenem-susceptible isolates Fig. 1 (b).

Global analyses of PDC variants from 19,478 *P. aeruginosa* genomes as obtained from NCBI were compared with the reference strains PAO1

that has PDC-1 allele. The prevalence of different PDC alleles was determined, revealing ESAC categories such as PDC-3 at 17.28 % (3367/19478) and PDC-5 at 12.91 % (2516/19478). Classical class C beta-lactamase categories included PDC-8 at 10.65 % (2076/19478) and PDC-35 at 7.42 % (1446/19478), while PDC-1 itself had a prevalence of 7.24 % (1412/19478). In our analysis of 131 *P. aeruginosa* isolates from India obtained from NCBI, the ESBL category PDC-3 was identified as the most prevalent allele, accounting for 19.84 % (26/131). Following



(a)

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      10      20      30      40      50      60      70
MRDTRFPCLC GIAASTLLFA TTPAAGEAP ADRLKALVDA AVQPVMKAND IPGLAVAIISL KGEPHYFSYG

      80      90     100     110     120     130     140
LASKEDGRRV TPETLFEIGS VSKTFTATLA GYALTQDKMR LDDRASQHPW ALQGSRFDDGI SLLDLATYTA

     150     160     170     180     190     200     210
GGLPLQFPDS VQKDQAQIRD YYRQWQPTYA PGSQRLYSNP SIGLFGYLA RSLGQPFER MEQQVFPALG

     220     230     240     250     260     270     280
LEQTHLDVPE AALAQYAQGY GKDDRPLRVG PGPLDAEGYG VKTSAADLLR FVDANLHPER LDRPWAQALD

     290     300     310     320     330     340     350
ATHRGYYKV DNTQGLGWSA YDWPISKRL QAGNSTPMAL QPHRIARLPA PQALEGQRL NKTGSENGIG

     360     370     380     390     397
AYVAFVPGRD LGLVILANRN YPNAERVKIA YAILSGLEQQ GKVPLKR

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(b)

Fig. 2. (a): Omega loop: Highlighted in red, spanning from amino acid position 178 to 226. R2 loop: Highlighted in green, spanning from amino acid position 289 to 307. H-11 Helix: Highlighted in yellow, extending from amino acid position 346 to the C-terminal end. Region interacting with C of Beta lactams: Highlighted in pink, located within the H-11 Helix, from amino acid position 346 to 349. (b): The protein sequence alignment of *P. aeruginosa* PDC-1 reveals several highly conserved regions, highlighted in red. These regions encompass S-V-S-K, Y-S-N, and K-T-G. Helix H2 is delineated by amino acids A97 and T105, both accentuated in bold and underlined. The omega loop is distinguished by yellow shading, while the R2 position stands out in green. Furthermore, the H11 helix is identified by pink highlighting. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

closely, PDC-5 was the second most common allele, present in 9.92 % (13/131) of the isolates. Classical beta-lactamase categories included PDC-11, PDC-19a, and PDC-8, with prevalence rates of 15.26 %, 14.50 %, and 6.10 %, respectively Fig. 1 (b).

4. Discussion

Evolution of PDC variants have been associated with the emergence of beta-lactam resistance, as evidenced by the reporting of over 500 allelic variants in global level. (<https://www.ncbi.nlm.nih.gov/pathogens/refgene/#PDC>). This diversity highlights the enzyme's remarkable polymorphism, capable of accommodating various amino acid substitutions, insertions, and deletions (Rodríguez-Martínez et al., 2009a). Recent investigations emphasize that clinical resistance to beta-lactams primarily stems from specific modifications within conserved motifs of PDC, implying mechanism for resistance acquisition. These modifications induce conformational changes that enhance the enzyme's catalytic efficiency, notably affecting regions such as the omega loop, R2 domain, and helix H region. Such alterations can give rise to extended-spectrum AmpCs (ESACs) with modified hydrolytic profiles and resistance to inhibitors (Chen et al., 2006; Hussein et al., 2024; Mora-Ochomogo and Lohans, 2021). Recently, we have identified two distinct PDC variants, namely blaPDC-546 (accession number WP_209359118) and blaPDC-547 (accession number WP_275802705) from CMC, Vellore. These variants have been found to impart resistance to cephalosporins, carbapenems, and beta-lactam inhibitors. Notably, both clinical strains exhibited complex antibiotic resistance profiles. PDC-546, characterized by the presence of K35E and T105A mutations, while, the PDC-547, harboring mutations G27D, T105A, V205L, G348D, and G391A.

PDC variants found in Indian isolates *P. aeruginosa*, those with an occurrence rate of 10 % or higher exhibit four prevalent amino acid substitutions: G27D, T105A, V205L, and G391A. These substitutions are consistently observed across PDC-447, PDC-398, and PDC-397 variants. These mutations suggested the emergence of PDC-447, PDC-398, and PDC-397 from PDC-3, as identified through protein BLAST analysis. Additionally, specific amino acid changes to these three PDC alleles P7A & S306T in PDC-447, V356T & N373I in PDC-397, and A97V & H323Q in PDC-398 are also identified. The amino acid sites 97 and 105 belong to the H2 helix (Fig. 2 (a, b)), and substitutions A97V and T105A may enhance the bacteria's capacity to hydrolyze antibiotics (4th Generation of cephalosporins) (Mammeri and Nordmann, 2007) (Raimondi et al., 2001). Specifically, the T105A alteration is believed to confer resistance against oxymino-cephalosporins and imipenem antibiotics in *P. aeruginosa* (Raimondi et al., 2001). Mutations V205L and P7S are associated with the Ω-loop and signal peptide, respectively, with the Ω-loop comprising amino acids from 178 to 226, and the signal peptide spanning amino acids 1 to 26 (<https://arpbigdisba.com/ps-eudomonas-aeruginosa-derived-cephalosporinase-pdc-database/>). The region from 289 to 307, forming the R2 loop, encompasses the S306T mutation in PDC-447. Substitutions V356I & N373I in PDC-397 are part of the region 346 to the C-terminal end, constituting the H11 helix of the enzyme (Nordmann and Mammeri, 2007). The H323Q mutation of PDC-398 is in proximity to the key catalytic residues Lys315 and Ala318 (Jacoby, 2009). PDC variants carried by strains, resulting from mutations in conserved regions, are known to confer resistance to antipseudomonal agents. A table illustrating reported PDC variants that are clinically significant, based on mutations located in the conserved region, can provide insights into their conferring resistance to BL/BLI (Table 1).

We have identified two inhibitor-resistant variants from this isolates PDC-224 and PDC-327. These variants have been reported as inhibitor-resistant variants in NCBI databases (NG_056102.1) and (NG_066534.1). They are commonly associated with specific mutations R5G, T21A, P23S, A36T, T105A, Q117L, A170T, L200I, V205L, E247K, R273K, V356I, and G391A. Additionally, the PDC-224 variant has a mutation

Table 1
Clinical significance of PDC variants: Mutations and positions in conserved regions conferring resistance to old and for new β-lactam and β-lactamase inhibitors.

PDC variants	Mutations	Position	Clinical Significance	References
PDC-2 PDC-3 PDC-5	G27D, A97V, T105A, V205L T105A R79Q, T105A	N-terminal, H2-helix, Ω-loop H-helix H-helix	Oxymino-cephalosporins and Imipenem	Rodríguez-Martínez et al., 2009b
PDC-3 PDC-315	T105A D245N	H-helix Ω-loop	Ceftolozane-tazobactam resistance	Arca-Suárez et al., 2020
PDC-3, PDC-8, PDC-24, PDC-35	E221K T105A, L176R T21A, T105A, G27D, A97V, T105A, V205L, G391A	Ω-loop H2-helix Signal peptide, H2-helix N-terminal, H2-helix, Ω-loop, C-terminal	Piperacillin-tazobactam resistance	Berrazeg et al., 2015
PDC-4	K108E		Cefepime resistance	Rodríguez-Martínez et al., 2009b
PDC-44 PDC-88	T105A, L320P G27D, T105A, V205L, ΔT316, ΔP317, G391A	H2-helix N-terminal, H2-helix, H11-helix, Ω-loop, C-terminal	Conferring higher resistance to ceftazidime, cefepime, and ceftolozane-tazobactam	Berrazeg et al., 2015
PDC-89 PDC-90 PDC-91 PDC-92	G391A G27D, T105A, V205L, ΔT316, ΔP317, ΔM318, G391A G27D, T105A, V205L, ΔT316, ΔP317, ΔM318, V356I, G391A G27D, T105A, V205L, ΔL320, ΔQ321, G391A			
PDC-221 PDC-223	Q219K ΔGly203 – E219	Ω-loop	The greatest increase in cephalosporin resistance as well as a marked reduction on piperacillin and imipenem MIC	Fraille-Ribot et al., 2020
PDC-477 PDC-482	A201T G214R	Ω-loop	Significance level of cephalosporin resistance, decrease piperacillin-tazobactam MIC	Ruedas-López et al., 2022
PDC-73 PDC-81 PDC-82	T105A, P180L T105A, P180L, G391A T105A, M201L, G391A	H2-helix, Ω-loop H2-helix, Ω-loop, C-terminal H2-helix, Ω-loop, C-terminal	Confer resistance to ceftazidime, ceftolozane-tazobactam	Berrazeg et al., 2015

(continued on next page)

Table 1 (continued)

PDC variants	Mutations	Position	Clinical Significance	References
PDC-84	V45A, T105A, V239A, N373I, G391A	H2-helix, C-terminal	Confer resistance to cephalosporins and carbapenem	Berrazeg et al., 2015
PDC-74	T105A, G242R	H2-helix, Ω-loop, C-terminal	Weakly antagonized by cloxacillin	Berrazeg et al., 2015
PDC-75	T105A, G242R, V356I			
PDC-78	R126H, G242R			
PDC-79	E247K			
PDC-80	E247G			
PDC-85	T105A, Y249H, G391A	H2-helix, C-terminal	High-level resistance to ceftolozane-tazobactam	Sid Ahmed et al., 2022
PDC-86	G27D, T105A, V205L, E247K, V356I, G391A	H2-helix, Ω-loop, C-terminal		
PDC-35	G27D, A97V, T105A, V205L, G391A	N-terminal, H2-helix, Ω-loop, C-terminal		
PDC-423	G183D	Ω-loop		
PDC-37	P7S, G27D, T105A, V205L, V356I, G391A	Signal peptide, N-terminal, H2-helix, Ω-loop, C-terminal	Resistance to ceftazidime, Cefepime, Aztreonam, Avibactam, Imipenem, Meropenem, Tazobactam	Lahiri et al., 2015
PDC-39	T105A, V205L, G391A	H2-helix, Ω-loop, C-terminal		
PDC-8	T105A, L176R	H2-helix, H5-helix		
PDC-9	T105A, Q155R, V205L	H2-helix, H5, Ω-loop		
PDC-10	T21A, T105A	Signal peptide, H2-helix		

G77S, and PDC-327 variant has a mutation G248S. These variants exhibit multiple mutations within the signal peptide and Ω loop regions, indicating potential resistance to beta-lactamase inhibitors. Moreover, PDC-3, PDC-5, PDC-8, PDC-11, PDC-19a are commonly associated with T105A substitution in the H2-helix conserved region.

The widespread presence of PDC variants in Indian isolates are not restricted solely to carbapenem resistant strains but are distributed across diverse susceptibility profiles. This suggests a broader adaptability of these variants across different resistance phenotypes, posing implications for understanding their role in antibiotic resistance mechanisms and treatments targeting *P. aeruginosa* infections.

Globally, PDC-3 emerges as the most prevalent variant at 17.28 %, characterized by a mutation in the helix-H region proximal to the omega loop position. Following closely, PDC-5 exhibits prevalence at 12.91 % in global isolates, indicating their notable occurrence among *P. aeruginosa* strains across various geographic regions. In the case of Indian isolates, the prevalence of PDC variants PDC-3 prevails at 19.84 %, followed by PDC-5 at 10 %, suggesting a distinct distribution pattern within Indian isolates compared to the global prevalence, which raises intriguing questions about the regional variations in strain prevalence. Indian isolates notably exhibit a higher prevalence of PDC-11 and PDC-19a, both present in more than 15 % of the isolates. This significant divergence in prevalence rates between global and Indian isolates underscores geographical disparities and adaptations of certain variants

within specific regions. The co-existence of predominant PDC variants in both carbapenem-resistant and susceptible strains challenges conventional notions linking specific variants exclusively to resistance. This exposure suggests a complex relationship between genetic variations in PDCs and their influence on diverse resistance profiles, necessitating a deeper exploration of these variants' functional implications. The substantial variance in prevalence between global and Indian isolates, particularly the higher occurrence of specific variants like PDC-3, PDC-5, PDC-11, PDC-19a and PDC-1 in Indian isolates, invites speculation about regional influences on strain prevalence and whether certain variants exhibit increased adaptability or prevalence within specific geographic regions (Fig. 3).

An in-depth exploration of identified mutations, particularly the T105A alteration, is essential to comprehend their potential impact on the catalytic properties. Most clinically significant PDC variants have emerged from the T105A substitution, bearing PDC-3 globally (Fig. 4). Understanding the prevalence patterns of these variants and their associations with resistance/susceptibility profiles holds paramount importance in formulating effective treatment strategies against *P. aeruginosa* infections. Furthermore, exploring the clinical implications of these findings regarding treatment regimens and potential novel therapies stands as a crucial step toward addressing antibiotic resistance and improving patient outcomes.

4.1. Further study/perspective

The increasing severity of *Pseudomonas aeruginosa* infections, driven by prolonged antibiotic use, has contributed to the evolution of strains that are difficult to treat (DTR), largely due to PDC variants. These evolved PDC variants are capable of hydrolyzing even 4th and 5th generation cephalosporins, rendering antipseudomonal drugs less effective. Future therapeutic strategies should prioritize targeting PDC-producing strains by focusing on mutations within highly conserved regions, such as the omega loop, R loop, and helix H, which are critical for binding affinity and catalytic properties. Research should concentrate on developing mutation-specific inhibitors and designing antibiotics that can inhibit PDC regulators, as downregulating regulator expression would reduce overall PDC expression and potentially restore antibiotic efficacy. Furthermore, regional surveillance of PDC variant distribution is essential to inform tailored interventions and effectively combat the evolving resistance mechanisms in *P. aeruginosa*.

Disclosure statement

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CRedit authorship contribution statement

Maruthan Karthik: Writing – original draft, Formal analysis, Conceptualization. **Srujal Kacha:** Methodology, Formal analysis, Data curation. **Subbulakshmi Rajendran:** Software, Methodology, Formal analysis, Data curation. **Yamuna Devi Bakthavatchalam:** Writing – review & editing, Validation, Supervision. **Binesh Lal:** Writing – review & editing, Validation, Supervision, Project administration. **Kamini Walia:** Writing – review & editing, Validation, Funding acquisition. **Balaji Veeraraghavan:** Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition, Conceptualization.

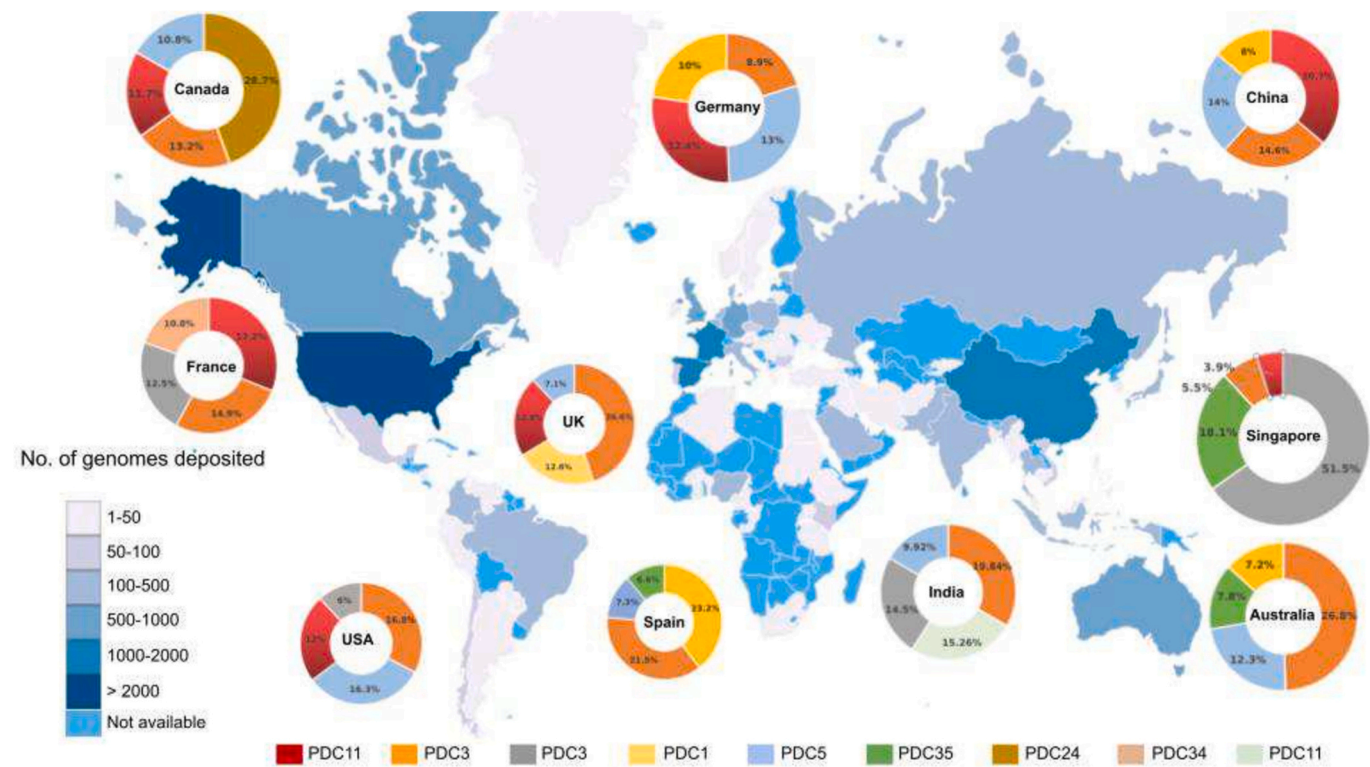


Fig. 3. The prevalence of PDC variants varies across countries, reflecting diverse genetic backgrounds and clinical significance.

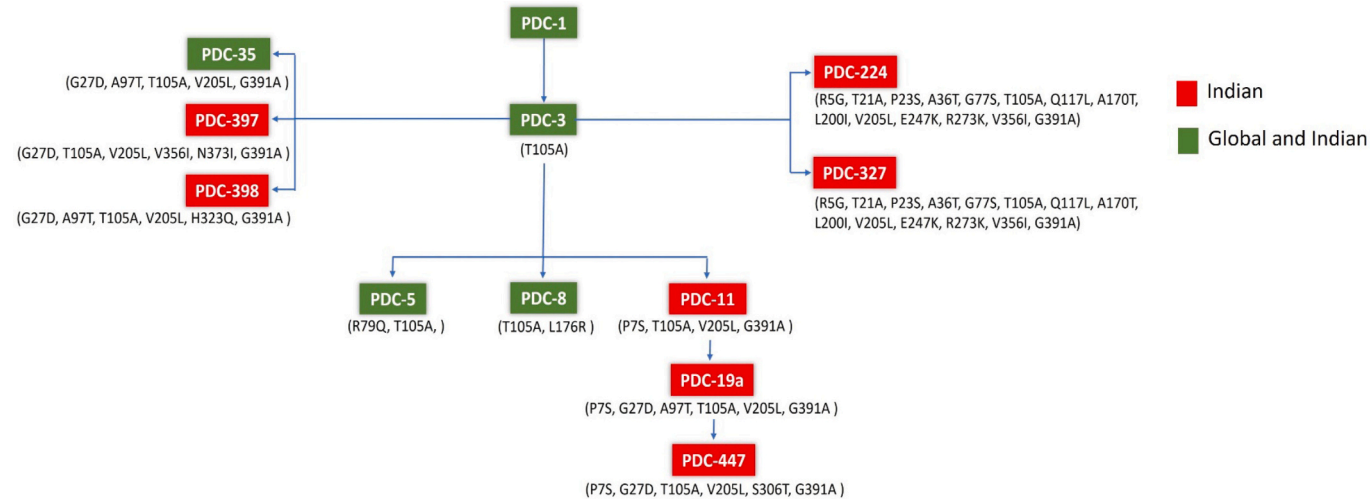


Fig. 4. Prevalence of predominant PDC variants, their evolution and distribution in global and Indian Isolates of *P. aeruginosa*.

Declaration of competing interest

Dr. Balaji Veeraraghavan reports financial support was provided by Christian Medical College Vellore. Dr. Balaji Veeraraghavan reports financial support was provided by Indian Council of Medical Research. Dr. Balaji Veeraraghavan reports a relationship with Christian Medical College Vellore that includes: employment. Dr. Balaji Veeraraghavan has patent pending to NA. NA If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.meegid.2024.105701>.

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